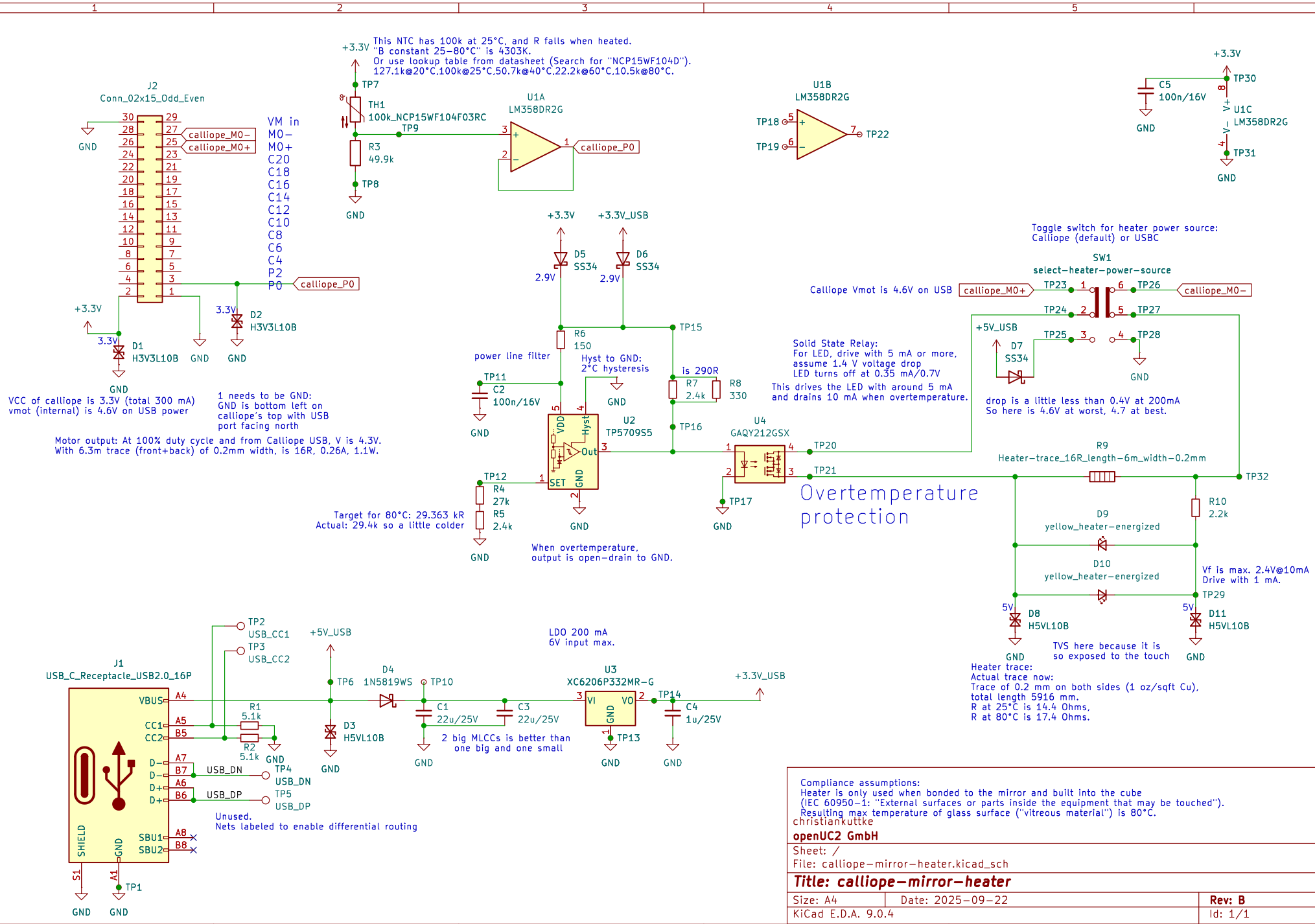
J2
Conn_02x15_Odd_Even
GND
calliope_M0−
calliope_M0+
VM in
M0−
M0+
C20
C18
C16
C14
C12
C10
C8
C6
C4
P2
P0
calliope_P0
+3.3V
3.3V
D1
H3V3L10B
D2
H3V3L10B
VCC of calliope is 3.3V (total 300 mA)
vmot (internal) is 4.6V on USB power
1 needs to be GND:
GND is bottom left on
calliope's top with USB
port facing north
Motor output: At 100% duty cycle and from Calliope USB, V is 4.3V.
With 6.3m trace (front+back) of 0.2mm width, is 16R, 0.26A, 1.1W.
This NTC has 100k at 25°C, and R falls when heated.
"B constant 25−80°C" is 4303K.
Or use lookup table from datasheet (Search for "NCP15WF104D").
127.1k@20°C,100k@25°C,50.7k@40°C,22.2k@60°C,10.5k@80°C.
TP7
TH1
100k_NCP15WF104F03RC
TP9
R3
49.9k
TP8
U1A
LM358DR2G
U1B
LM358DR2G
TP18
TP19
TP22
C5
100n/16V
U1C
LM358DR2G
TP30
TP31
+3.3V_USB
D5
SS34
D6
SS34
2.9V
R6
150
power line filter
TP11
C2
100n/16V
Hyst to GND:
2°C hysteresis
TP15
is 290R
R7
2.4k
R8
330
U2
TP5709S5
TP16
U4
GAQY212GSX
TP20
TP21
TP17
TP12
R4
27k
R5
2.4k
Target for 80°C: 29.363 kR
Actual: 29.4k so a little colder
When overtemperature,
output is open−drain to GND.
Toggle switch for heater power source:
Calliope (default) or USBC
SW1
select−heater−power−source
Calliope Vmot is 4.6V on USB
TP23
TP24
TP25
TP26
TP27
TP28
+5V_USB
D7
SS34
drop is a little less than 0.4V at 200mA
So here is 4.6V at worst, 4.7 at best.
Solid State Relay:
For LED, drive with 5 mA or more,
assume 1.4 V voltage drop
LED turns off at 0.35 mA/0.7V
This drives the LED with around 5 mA
and drains 10 mA when overtemperature.
Overtemperature
protection
R9
Heater−trace_16R_length−6m_width−0.2mm
TP32
R10
2.2k
D9
yellow_heater−energized
D10
yellow_heater−energized
Vf is max. 2.4V@10mA
Drive with 1 mA.
TP29
D8
H5VL10B
D11
H5VL10B
TVS here because it is
so exposed to the touch
Heater trace:
Actual trace now:
Trace of 0.2 mm on both sides (1 oz/sqft Cu),
total length 5916 mm.
R at 25°C is 14.4 Ohms,
R at 80°C is 17.4 Ohms.
J1
USB_C_Receptacle_USB2.0_16P
TP2
USB_CC1
TP3
USB_CC2
R1
5.1k
R2
5.1k
USB_DN
USB_DP
TP4
USB_DN
TP5
USB_DP
Unused.
Nets labeled to enable differential routing
+5V_USB
TP6
D4
1N5819WS
TP10
D3
H5VL10B
LDO 200 mA
6V input max.
U3
XC6206P332MR−G
C1
22u/25V
C3
22u/25V
2 big MLCCs is better than
one big and one small
TP13
TP14
C4
1u/25V
+3.3V_USB
TP1
Compliance assumptions:
Heater is only used when bonded to the mirror and built into the cube
(IEC 60950−1: "External surfaces or parts inside the equipment that may be touched").
Resulting max temperature of glass surface ("vitreous material") is 80°C.
christiankuttke
openUC2 GmbH
Sheet: /
File: calliope−mirror−heater.kicad_sch
Title: calliope−mirror−heater
Size: A4
Date: 2025−09−22
Rev: B
KiCad E.D.A. 9.0.4
Id: 1/1